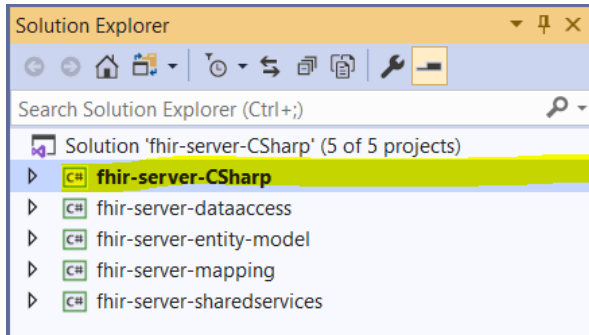


FHIR Server Implementation Assignment

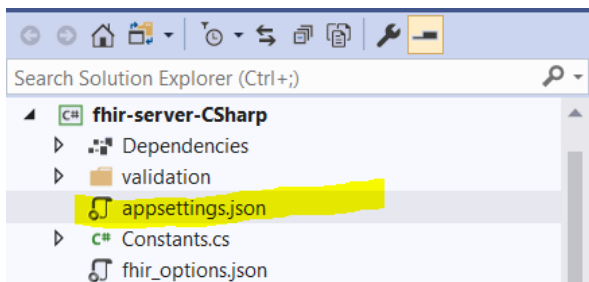
C# server implementation – Operations Guide

Note: - I am not familiar with Java or NodeJS at all. Therefore, I implemented a server which is a basic http listener which will cater for Patient, Practitioner and Medication Request profiles.

1. Set Start up project as “**fhir-server-CSharp**” by right-clicking the project and select “Set As Startup Project”

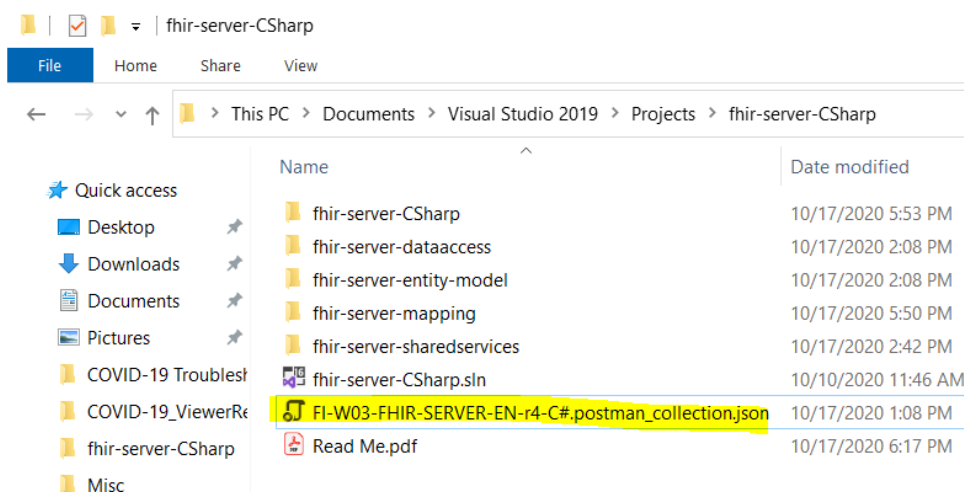


2. Listener is listening on port **5834** for any incoming HTTP traffic. If your machine is using this port, change it to a different port. You can change this by modifying **appsettings.json** configuration file which can be found in “**fhir-server-CSharp**” project.



```
    "**/obj",  
    "**/platforms"  
  ],  
  "ListenerConfig": {  
    "PortListening": 5834,  
    "FHIRServerUrl": "/api/FHIR/"  
  },  
  ...  
}
```

3. A CapabilityStatement has implemented and that defines all operations which can be performed against this server.
4. I bundled all the queries in “**FI-W03-FHIR-SERVER-EN-r4-C#.postman_collection.json**” postman file, so it is easy for you to access the functionalities of this server.



- This server supports POST MedicationRequest JSON data validation by utilizing the specification defined for US Core profiles. Nuget library has HL7.Fhir.Specification.R4 library for this and this server uses this library. This feature cannot be turned off because it is a MUST to validate user supplied data to make sure that the JSON payload contains proper cardinalities defined in US Core MedicationRequest profile (<https://www.hl7.org/fhir/us/core/StructureDefinition-us-core-medicationrequest.html>).

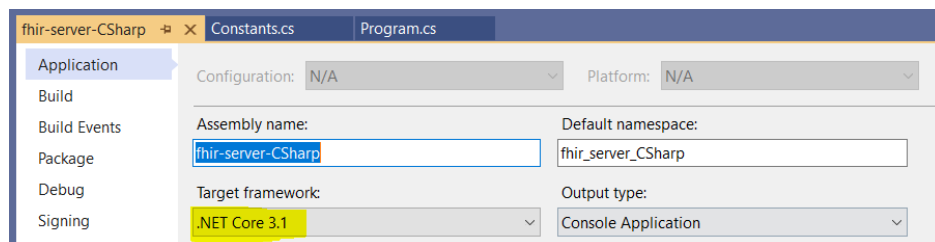
Also, GET validates bundles and resources as well. As the output profile validation takes some time, I disabled that feature. You can enable the feature by modifying the **appsettings.config** and the following properties.

```
},|
"ResourceValidation": {
  "ValidatePractitionerBundleAndResource": false,
  "ValidatePatientBundleAndResource": false,
  "ValidateMedicationRequestBundleAndResource": false,
  "ValidateCapabilityStatementResource": false
}
```

- Database has new tables for Medication Request and two new views. Please be sure to use the database provided with this solution.

Name	Type	Schema
Tables (7)		
DOC_TYPE	CREATE TABLE "DOC_TYPE" ("DC"	
MED_REQ_DETAILS	CREATE TABLE "MED_REQ_DETAIL:"	
MED_REQ_INTENT_CODES	CREATE TABLE "MED_REQ_INTENT	
MED_REQ_STAT_CODES	CREATE TABLE "MED_REQ_STAT_C	
PERSON	CREATE TABLE "PERSON" ("PRSN_	
PERSON_DOC	CREATE TABLE "PERSON_DOC" ("I	
sqlite_sequence	CREATE TABLE sqlite_sequence(na	
Indices (0)		
Views (2)		
v_fhir_server_GetAllPeople	CREATE VIEW v_fhir_server_GetAllIF	
v_fhir_server_GetAllPractitioners	CREATE VIEW v_fhir_server_GetAllIF	
Triggers (0)		

- This project is built using .NET Core framework 3.1 and Visual Studio 2019 is used as the IDE for all projects.



Visual Studio

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Microsoft Visual Studio Professional 2019
Version 16.7.6

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Microsoft .NET Framework
Version 4.8.04084

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8. PERSON, DOC_TYPE and PERSON_DOC table structures never changed as per the assignment requirement and I leverage the existing database/table structure to load Practitioners as well. As per the capability statement, Patient and Practitioner resources only supports READ. So if you wish to create new Patient and Practitioner resources, you could copy this database to NODE JS or JAVA project and run the application, create the resources as you want and then get the new database and replace the one which I have attached with this solution. If you wish to have POST feature for Practitioner and Patient in C# server, please let me know and I will put some time to implement that feature too.
9. Finally, I put some real effort on creating this server using C# and I UNIT tested to eliminate all possible errors. I think I fulfilled all requirements asked in this assignment too. If you find any difficulties running this project please let me know and I will provide 100% support to run the project. You can contact me via Chandima.fonseka@ontariohealth.ca or Lakmal.asia@gmail.com.